

# AR Protein Dynamics Installation Guide

Wake Tech Community College

CSC 492 Team 8

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# Introduction

Our project can be configured to run in two different environments: on a local development machine, and on a remote server:

- Running on a local machine should be used for development purposes.
- When testing the application on a mobile device or for demonstration purposes the project can be run on a remote server.

## Install on Local Machine

### STEP 1: Install Docker on Your System

1. Using the official Docker documentation, install Docker on your machine. Docker will be used to isolate the 4 main components of the system, including NGINX, Frontend, Backend, and Database.
2. Installation Link: <https://docs.docker.com/get-docker/>
3. Docker Desktop is useful to have as well, but not necessary:
  - a. <https://www.docker.com/products/docker-desktop/>

### STEP 2: Clone the GitHub Repository

1. Clone the GitHub repository onto your machine.
2. Repository Link: <https://github.ncsu.edu/engr-csc-sdc/2022FallTeam08-WTCC>

### STEP 3: Install Gradle (for Unit Testing)

1. For testing purposes, install Gradle on your machine via terminal in your developing environment. Follow each step in the link below.
2. Installation Link: <https://gradle.org/install/>

### STEP 4: Install MySQL Server

1. Using the MySQL Community documentation, install MySQL Server on your machine.
2. Installation Link: <https://dev.mysql.com/downloads/mysql/>
3. Using MySQL Server, create a server host with TCP/IP port

### STEP 5: Install MySQL Workbench

1. Using the official MySQL documentation, install MySQL on your machine. MySQL will be used to visualize the database.
2. Installation Link: <https://www.mysql.com/products/workbench/>
3. Once Installed - Setup a new connection
  - a. Username: 'root'
  - b. Hostname: *Insert Name or IP address of the server host*

- c. Port name: *TCP/IP port*

#### **STEP 6: Set up application.properties**

1. Create file called application.properties in server folder
2. Copy and paste the entire application.properties.template contents into application.properties
3. Set “spring.datasource.username” and “spring.datasource.password” equal to “root” for both
4. Replace TODO with whatever string your team decides on for “ar.app.jwtSecret”

#### **STEP 7: Start the Docker Compose**

1. Run the following command from the root directory of your git repository to start the system in development mode:
  - a. `docker-compose -f docker-compose.yml -f docker-compose.dev.yml up --build`
2. Run the following command from the root directory of your git repository to start the system in production mode:
  - a. `docker-compose -f docker-compose.yml -f docker-compose.prod.yml up --build`
3. Run either of the following commands with a -d at the end to run in detached mode

\* Before starting step 7, make sure:

- you are connected to the NCSU wifi or NCSU VPN

# Installing on Server

Most development will be done locally but this will be useful for testing your application on your phone.

## **STEP 1: Acquire a System to Run the Service**

This system can be acquired through a variety of means, but our team opted to use a virtual machine through NCSU's IT department.

1. We recommend that your machine runs a Linux distribution. Our virtual machine had the following specifications:
  - Operating System: Ubuntu Linux (Server)
  - CPU: 2 vCPU
  - RAM: 2GB
  - Storage: 40GB HDD
2. When requesting the virtual machine ask for:
  - docker to be installed
  - sudo privileges
  - ssl certificate
3. As the system is designed using docker compose, you could also opt to run the different services on different physical machines. While this is an option, it is quite convenient to run on a single machine if possible.

## **STEP 2: Connect to the virtual machine via SSH**

1. Open up a terminal (one where ssh is installed in the system)
  - a. to check type into terminal: ssh
  - b. ensure no errors pop up
2. Run the command: ssh unityld@virtual-machine-link-provided-by-IT
  - a. the password is you unityld password
  - b. you will not be proceeding with the steps in your Virtual Machine

## **STEP 3: Make Sure that Docker is installed on Your System**

1. Check if docker is installed on your virtual machine using the command: "sudo docker --version"
2. If docker is not installed already contact CSC IT and request for it to be installed
3. Installation Link: <https://docs.docker.com/get-docker/>

## **STEP 4: Clone the GitHub Repository in you Virtual Machine**

1. <https://github.ncsu.edu/engr-csc-sdc/2022FallTeam08-WTCC>

**STEP 5: Set up application.properties in project in Virtual Machine**

1. Create file called application.properties in server folder
2. Copy and paste the entire application.properties.template contents into application.properties
3. Set "spring.datasource.username" and "spring.datasource.password" equal to "root" for both
4. Replace TODO with whatever string your team decides on for "ar.app.jwtSecret"

**STEP 6: Set up nginx**

1. CSC IT should have provided you with SSL certificates and keys in your request for a VM and provided you the path with the location of the certs.
2. Then in nginx/prod/conf.d/default.conf, on lines 15 and 16, change the paths to the appropriate locations for the certificates/keys.
3. The docker-compose for docker-prod would work at this point.

**STEP 6: Start the Docker Compose**

1. Run the following command from the root directory of your git repository to start the system in production mode:
  - a. `sudo docker-compose -f docker-compose.yml -f docker-compose.prod.yml up --build`  
OR
  - b. `sudo make prod`
2. Run the command with a -d at the end to run in detached mode, you will need to run in detached mode if you want to keep it running and exit the server

\* If any issues arise with setting up the project on a server, contact NCSU CSC IT